CSE4065 – Fall 2022
Computational Genomics
Lecture 1: Introduction

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Agenda

- Course Info
- Course Outline
- Introduction to Computational Genomics

Course Info:

- Instructor: Asst. Prof. Betül Boz
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- Prerequisites
  - None
- Lecture Times:
  - Tuesday 14:00-15:00 M2.Z07
  - Wednesday 13:00-15:00 M2.Z07
- Main Text Book
  - Compeau Philip, Pevzner, Pavel: Bioinformatics Algorithms: An active Learning Approach
  - Jones and Pevzner: An Introduction to Bioinformatics Algorithms
  - Durbin, Eddy, Krogh, Mitchison: Biological Sequence Analysis
  - Makinen, Belazzougui, Cunial, Tomescu: Genome-Scale Algorithm Design

How Will You Be Evaluated? (Tentative)

- Quiz %10
- Assignment %25
- Midterm Exam %25
- Final Exam %40

Policies

- No late assignments accepted
- Everything must be your own work (unless otherwise specified)
- Cheating → Failing grade

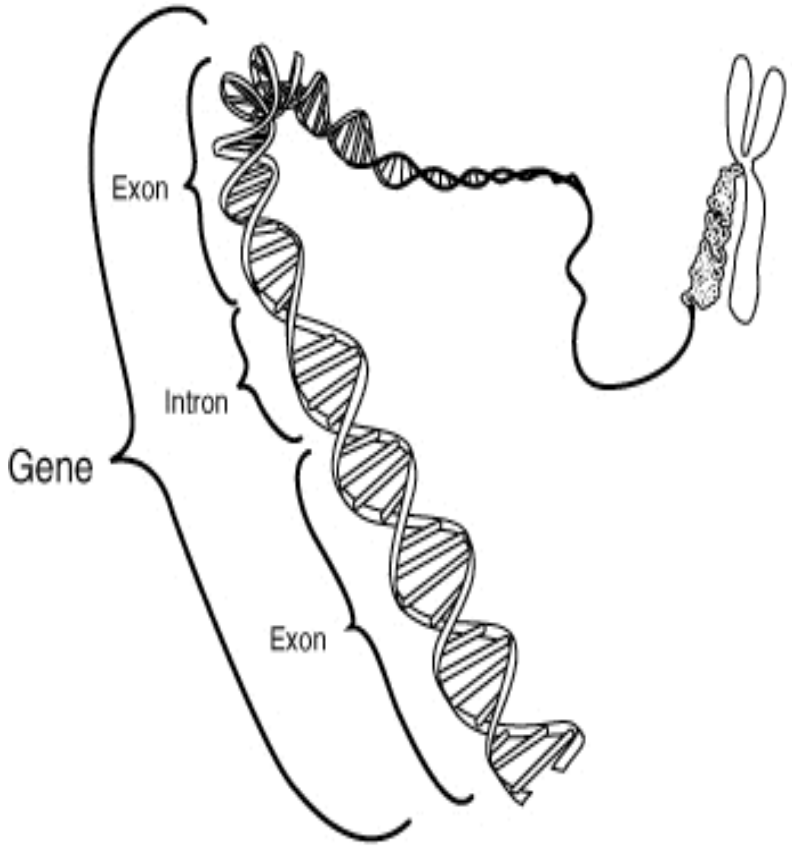
Course Outline

- DNA-RNA-Protein Structure
- DNA Replication
- DNA Patterns
- Motif Search
- Genome Assemble
- Sequence Allignment
- Multiple Sequence Alignment
- Genome Rearrangements
- Gene Prediction

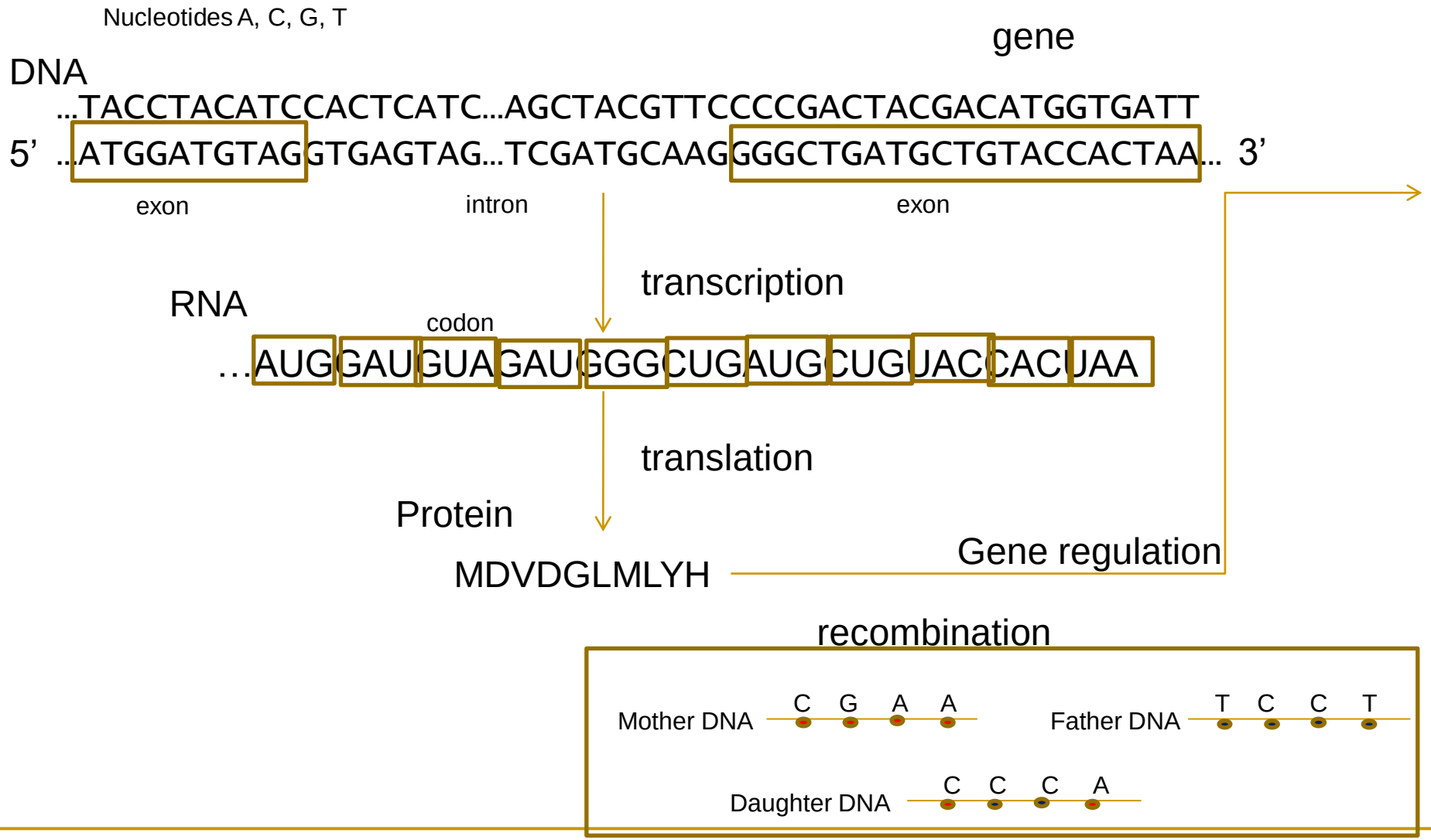
Genes

- Segment that specifies the sequence of a protein
  - Exons = coding sequences
  - Introns = non-coding sequences

- Occupies a specific location on a chromosome (an organized strand of DNA)



Molecular biology



What is Bioinformatics?

- Bioinformatics, *n.* The science of **information and information flow in biological systems**, esp. of the use of computational methods in **genetics and genomics**. (Oxford English Dictionary)

- "The **mathematical**, **statistical** and **computing** methods that aim to solve biological problems using **DNA and amino acid sequences** and related information." -- Fredj Tekaia

Why is bioinformatics important?

- New measurement techniques produce huge quantities of biological data
  - Advanced data analysis methods are needed to make sense of the data

- Paradigm shift in biology to utilise bioinformatics in research

Biological Sequence Analysis

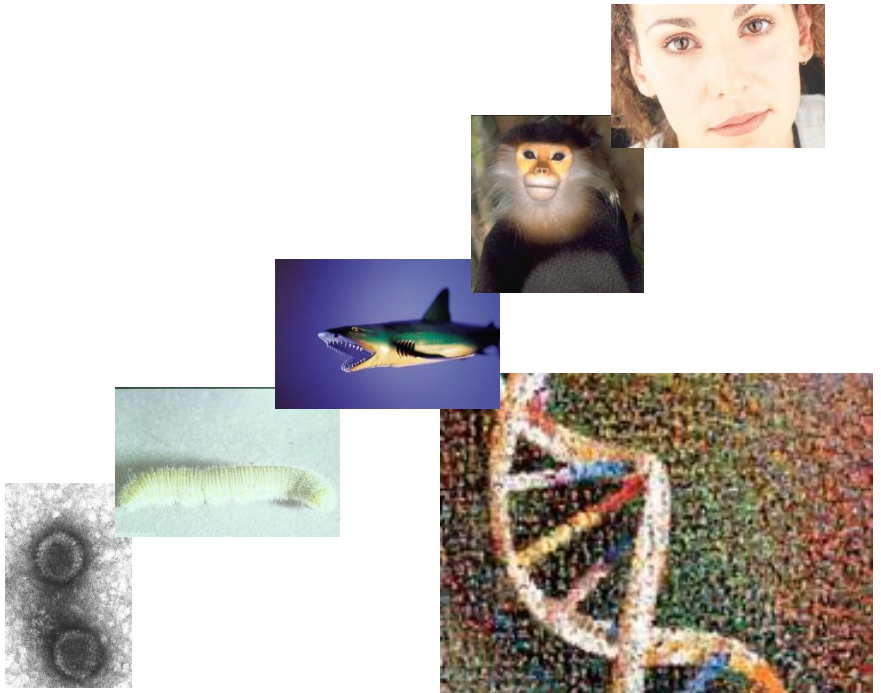
- DNA, RNA, and protein sequences are the fundamental level of biological information.
- Analysis of such biological sequences forms the backbone of all bioinformatics.

Scientific Method of Bioinformatics

- Bioinformatics is not a science in itself, just a new approach to study a science – biology.
- The accepted way to do research in bioinformatics is somewhere between the hypothesis testing method of experimental sciences and exact mathematical method of exact sciences.
  - There are two extremes among bioinformaticians:
    - Those that use bioinformatics tools in creative ways and follow the hypothesis testing method of experimental sciences.
    - Those that develop the bioinformatics tools and follow the exact mathematical method.
  - Typically the most influental research is done somewhere in between.

Computational Genomics

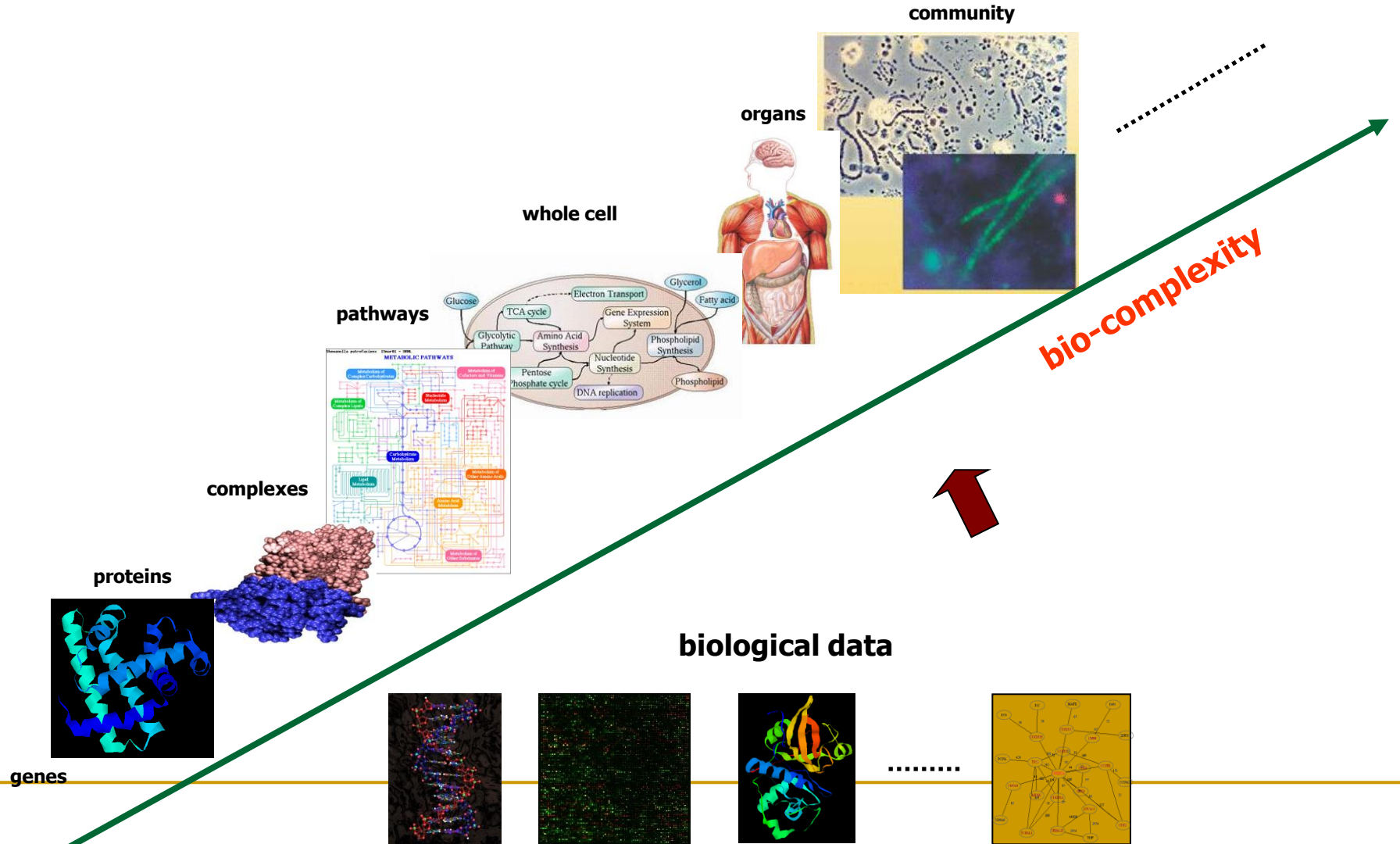
- Our genome encodes an enormous amount of information about our beings
  - our looks
  - our size
  - how our bodies work
  - ….
  - our health
  - our behaviors
  - … who we are!



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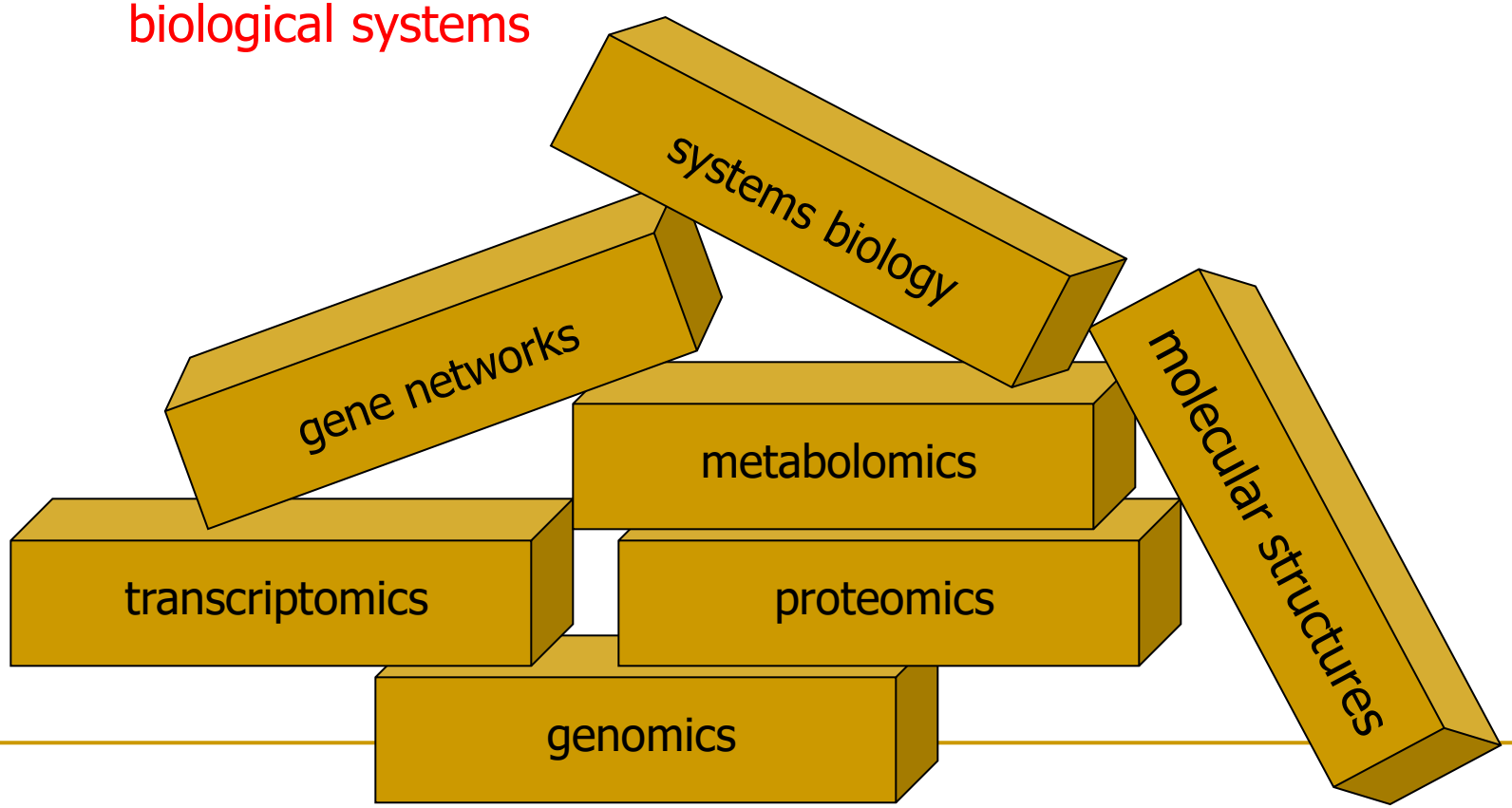
Computational Genomics

- As technologies improve, we are able to extract more and more information encoded in a genome



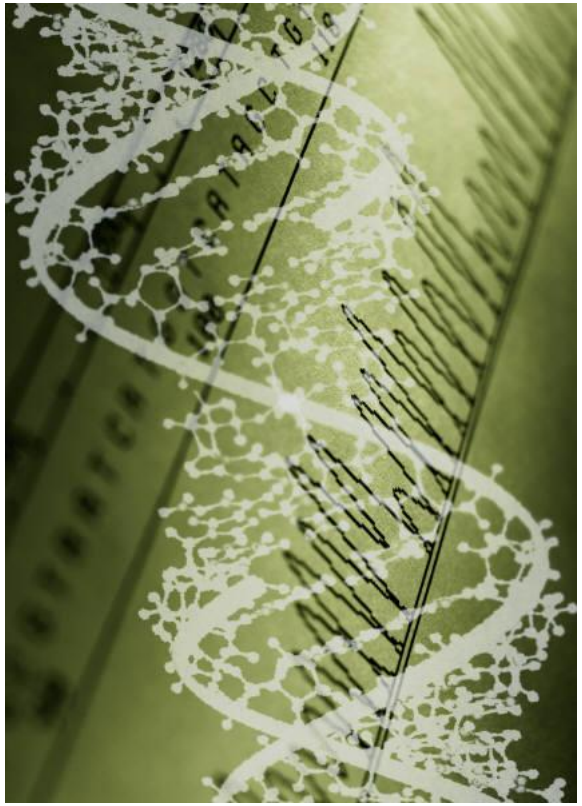
Computational Genomics

- While the ultimate goal of "functional genomics" is to link behavior of cells, organisms, and populations to the information encoded in the genome, "computational genomics" is mainly about identifying and characterizing the parts-lists of complex biological systems



Computational Genomics

- Genetic parts-list encoded in a genome
  - genome sequence and variations
  - genomic structures
  - protein-coding genes
  - RNA-coding genes
  - pseudo genes
  - homologs/orthologs/paralogs
  - promoters/terminators
  - regulatory elements/binding motifs
  - transposable elements
  - .....



Computational Genomics

- Computational genomics is playing an increasingly more important role in modern biology
  - suggesting possible genes associated with a particular disease, and hence reducing the search space for relevant genes
    - e.g., genes involved in retinal disease
  - suggesting an organism's biology through genome comparison,
  - providing powerful tools for studies of biological evolution
    - sequence/genome comparison
  - have played key roles in the human and other genome projects
    - protein-coding gene prediction

Challenges in Computational Genomics

- Prediction of protein-coding genes still represents a challenging problem
  - accurate prediction of exon/intron boundaries
  - prediction of alternatively spliced gene forms

- Protein-coding genes account for ~3% of the human genome. What and where are the other "functional elements" in the rest of the genome?
  - how to identify them?
  - how to (help to) predict their functions?

Challenges in Computational Genomics

- Identification of RNA-coding genes
  - what are the identifiable characteristics of RNA genes?
- Particularly, identification of small regulatory RNA
  - short interference RNAs (siRNA)
  - microRNA (miRNA)
  - small modulatory RNA (smRNA)
- Genome is not a linear sequence; It is a 3D structure!
  - accurate identification and characterization of functional elements by looking at the genome as a 3D DNA structure